Industrial-Sentinel: Autonomous Edge-AI for the Modern Factory

Protecting People, Products, and Profits with Agentic Intelligence.

The Problem (The "Cost of Chaos")

Nuisance Tripping: Traditional sensors stop the belt for a stray box, costing $500+/hour in downtime.

Human Error: Manual safety logs are often inaccurate or incomplete.

Environmental Drift: Karachi’s heat causes seal failures that lead to product spoilage (leakers).

Safety Hazards: Inconsistent "Hard Stops" can damage expensive conveyor motors.

One Edge-Hub, Four Intelligent Agents

Safety Agent: Uses YOLOv8 to distinguish "Legal Objects" (Cartons) from "Hazards" (Hands/Tools).

Compliance Agent: OCR-driven real-time label and expiry verification.

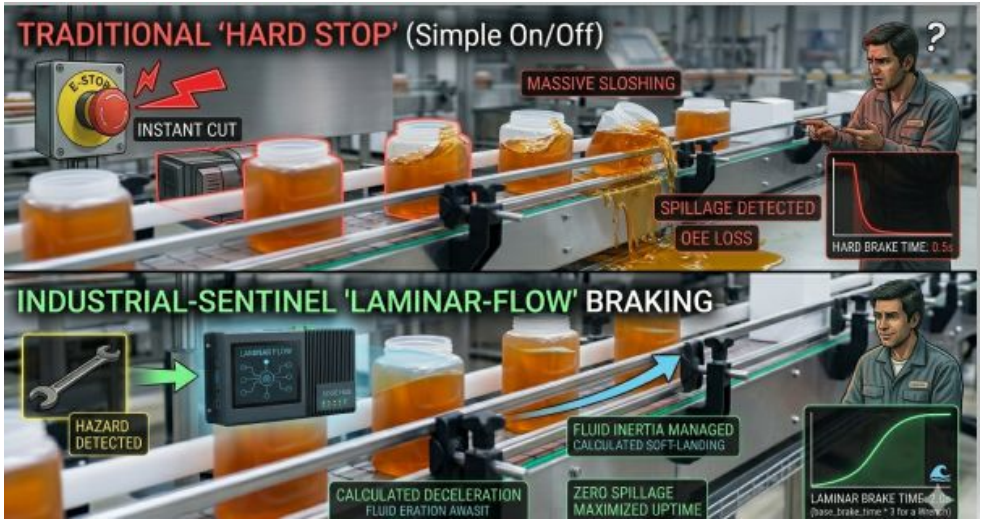
Thermal Agent: LangGraph-driven reasoning to compensate for ambient heat spikes.

Audit Agent: Automated, tamper-proof PDF reporting for every shift. **Speaker Notes:** "We run all of this locally on standard hardware. No expensive cloud subscriptions, no data leaks, and 100% uptime during internet outages."

Beyond Simple On/Off: Intelligent Deceleration

Key Feature: Laminar-Flow Braking

- **The Logic:** If the line carries liquids (Milk/Soap/Juice), a sudden stop causes massive spillage.
- **The AI Action:** Our agent calculates a 3x "Soft-Landing" braking profile.
- **Result:** Hazard is cleared, but the product stays inside the container. No mess, no waste.



100% Audit-Ready, 100% of the Time

Instant Accountability: Every "Emergency Stop" is timestamped and categorized.

Reduced Liability: Digital proof of safety protocols for insurance and legal audits.

Management Visibility: Executive summaries delivered to your inbox at the end of every shift.

High-End Intelligence on Everyday Hardware

Low CapEx: Runs on standard i5/i7 industrial PCs (No $5,000 GPU required).

Privacy First: Data stays on the factory floor.

Rapid Deployment: Modular architecture allows for setup in days, not months.

Secure Your Production Future

"Let’s start with a Pilot Module on a single line. We’ll prove the reduction in false-positives and the accuracy of our Thermal-Sentry within 30 days."